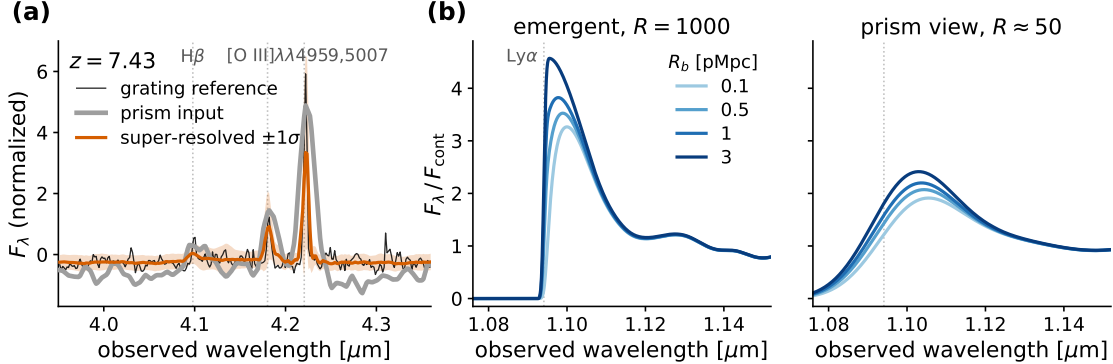


Deep Learning for Galaxy Spectral Super-Resolution

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Modern infrared telescopes like JWST and Nancy Grace Roman enable cosmic observations, but efficient spectroscopic modes operate at low resolution: JWST’s NIRSpec prism ($R \sim 100$) and Roman’s grism. Spectral features encoding critical physics blur below instrumental limits. I develop physics-informed deep learning frameworks that super-resolve low-resolution galaxy spectra by $\sim 10\times$, recovering hidden astrophysical information.

Using paired JWST/NIRSpec observations from JADES (same galaxies in high-resolution and prism modes), I train generative models with physics-informed loss functions enforcing accurate emission line ratios. Figure 1(a) demonstrates super-resolution on a $z = 7.43$ galaxy, recovering features invisible in prism data. Figure 1(b) shows the astrophysical impact: Lyman-alpha profiles encode ionized bubble sizes (R_b) around galaxies. High-resolution profiles are distinguishable; prism-resolution profiles appear identical. Our method recovers this hidden information, enabling physical measurements from scientifically limited data.



Summer 2027 at IPAC: The fellowship focuses on four integrated components: (1) validating recovery of astrophysical diagnostics using held-out JADES data; (2) adapting the framework to Roman grism with IPAC’s mission team; (3) developing multi-modal models integrating photometric and spectroscopic information; (4) extending methodology to Euclid spectroscopy. The timing is critical: Roman’s first data release is expected Summer 2027. On-site access to real mission data and Dr. Shoubaneh Hemmati—active member of Euclid and Roman missions—enables immediate validation and adaptation impossible remotely. In-person collaboration ensures our methodology aligns with actual mission requirements and delivers forecasts to inform survey strategy.

This work advances galaxy astrophysics and demonstrates how machine learning extracts maximum value from observational data. Open-source code, pre-trained models, and value-added catalogs will enable researchers to apply spectral super-resolution across datasets, extending the science reach of JWST, Roman, Euclid, and future missions.